

Project Proposal: Probabilistic Forecasting of Bike-Sharing Demand Under Weather and Seasonal Effects

Research question and data

This project asks: How well can hourly bike-sharing demand be predicted from weather, calendar, and seasonal variables, and does a probabilistic Bayesian model provide more useful uncertainty estimates than standard point-prediction approaches? Public bike-sharing datasets include hourly rental counts together with temperature, humidity, wind speed, season, holiday, and working-day indicators, making them suitable for a real-data forecasting study. [cite:197][cite:187]

The proposed data source is the publicly available UCI Bike Sharing Dataset, which contains hourly and daily rental counts from the Capital Bikeshare system together with weather and calendar covariates.[cite:197] This dataset is attractive because it is real, structured, and immediately relevant to operational planning: bike-sharing systems need accurate demand forecasts to position bicycles, allocate staff, and avoid shortages during peaks.[cite:181][cite:184]

Motivation and relevance

Bike-sharing is an important component of urban mobility and sustainability, especially in cities that want to reduce congestion and emissions.[cite:184][cite:190] Better demand forecasts help operators decide where to rebalance bikes, how many bikes to keep available, and when weather-driven demand shifts are likely to occur.[cite:181][cite:187]

The question is relevant for transportation planning, smart-city operations, and mobility-as-a-service platforms.[cite:184][cite:190] It also has a broader methodological relevance because it combines time patterns, external covariates, and uncertainty quantification, which are central themes in modern applied statistics.[cite:182][cite:187]

Past work and shortcomings

Past work on bike-sharing demand has often focused on deterministic machine-learning models such as linear regression, decision trees, random forests, gradient boosting, and XGBoost, with the main evaluation criterion being predictive accuracy.[cite:181][cite:184] These methods can perform well, but they typically emphasize point forecasts and may provide limited information about forecast uncertainty, which is crucial when planners must prepare for both typical and high-demand scenarios.[cite:184][cite:187]

Related transportation research also shows that temporal dependence and hierarchical structure matter; Bayesian hierarchical and autoregressive models have improved fit in crash-frequency applications compared with simpler Poisson models.[cite:182][cite:185] This suggests a shortcoming of many simple forecasting approaches: they may miss temporal structure or

understate uncertainty, especially when observations are counts influenced by weather and seasonality.[cite:182][cite:187]

Solution sketch and innovation

The proposed solution is a Bayesian count-regression model for hourly bike demand, potentially starting with a Poisson model and extending to a negative-binomial variant if overdispersion is present. Count distributions are appropriate because the response variable is the number of rentals in a fixed time interval, and overdispersion is plausible when demand spikes occur due to commuting peaks, weather shocks, or events.[cite:181][cite:182]

The model will include predictors such as temperature, humidity, wind speed, season, hour of day, weekday/holiday indicators, and possibly interaction terms between weather and season.[cite:181][cite:187] Tools from an introductory Bayesian workflow, such as prior specification, posterior inference, posterior predictive checks, and model comparison, will be used to estimate both expected demand and uncertainty bands.[cite:182][cite:187]

The innovation is not necessarily a completely new algorithm, but an applied comparison that emphasizes **probabilistic usefulness** rather than only average prediction error.[cite:187] Compared with standard point-prediction models, a Bayesian model can produce predictive intervals, make uncertainty explicit, and better support operational decisions such as whether extra bikes should be moved preemptively when bad weather or rush-hour conditions create volatile demand.[cite:182][cite:187]

Expected outcome and working hypothesis

The working hypothesis is that weather and calendar effects explain a substantial part of hourly variation, with temperature having a positive effect up to a point and adverse weather reducing demand.[cite:181][cite:190] Clear daily and seasonal patterns are also expected, especially commuter peaks on working days and lower demand during colder periods.[cite:181][cite:187]

The expected result is that a Bayesian model with appropriate count distribution and temporal predictors will deliver competitive forecasts while also giving informative uncertainty intervals that are more useful for real-world planning than single-number predictions alone.[cite:182][cite:187] Even if the mean prediction accuracy is only similar to standard machine-learning baselines, the probabilistic output is expected to be a practical advantage for fleet rebalancing and resource planning.[cite:184][cite:187]